Name: Javaughn Stevens

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MATLAB 2

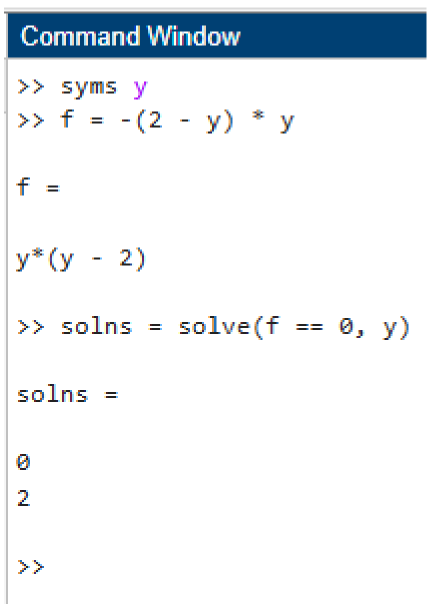
Exercise (B. Hunt)

Consider the critical threshold model for population growth $y' = -(2 - y)y$.

a. Find the stationary solutions (equilibrium solutions) of the differential equation.

We can rewrite the equation: dy/dt = -(2 - y)y.

Command Window Screenshot



```
Command Window
>> syms y
>> f = -(2 - y) * y

f =

y*(y - 2)

>> solns = solve(f == 0, y)

solns =

0
2

>>
```

% The stationary solutions are y = 0 and y = 2.

b. Draw the slope field and use it to decide which equilibrium solutions are stable and which are unstable ($y(t)$ is stable if solutions that start near $y(t)$ converge to $y(t)$).

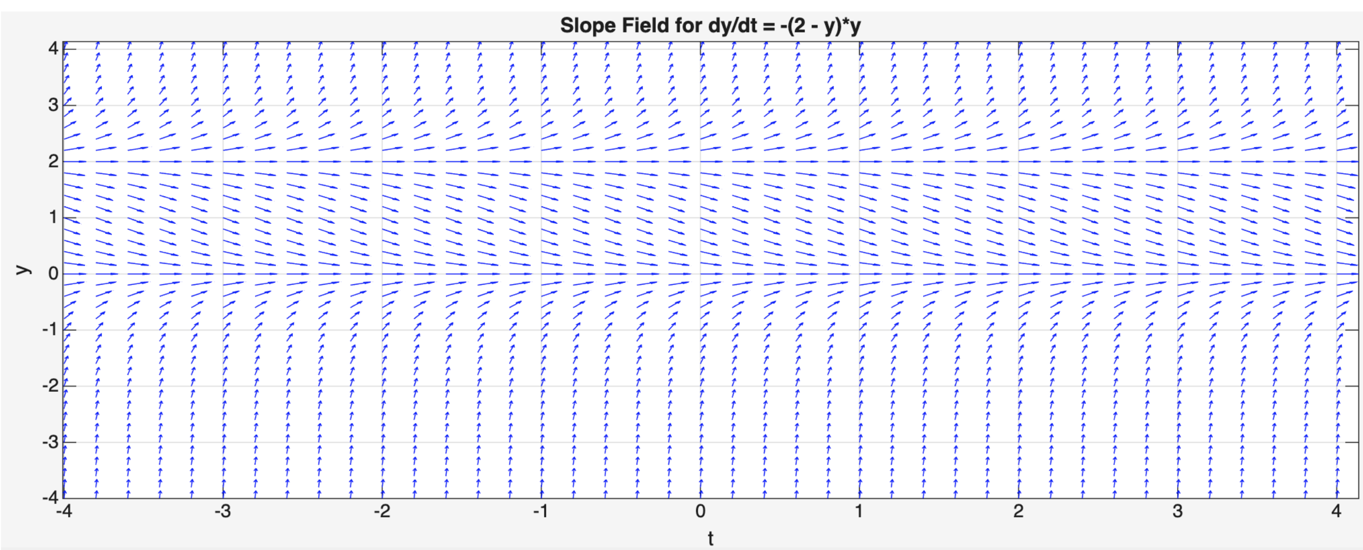
Hint: modify [T,Y]=meshgrid(-4:0.2:4, -4:0.2:4) until you obtain a desirable slope field.

Command Window Screenshot

Command Window

```
>> [T, Y] = meshgrid(-4:0.2:4, -4:0.2:4);
>> dY = -(2 - Y).*Y;
>> A = sqrt(1 + dY.^2);
>> B = 1 ./ A;
>> C = dY ./ A;
>> figure;
>> quiver(T, Y, B, C, 0.5, 'b');
>> axis tight
>> grid on
>> xlabel('t');
>> ylabel('y');
>> title('Slope Field for dy/dt = -(2 - y)*y');
```

Figure 1



% According to Figure 1, it appears that y = 0 is a stable solution and y = 2 is an unstable solution.

c. What is the limiting behavior of the solution if the initial population is between 0 and 2? Greater than 2?

% If 0 < y(0) < 2, dy/dt = - (2 - y)y **<** *0, so the population is <u>decreasing</u>, and as t -> ∞, y(t) will approach y = 0. The limiting behavior is y(t) -> 0.*

% If y(0) > 2, dy/dt = - (2 - y)y **>** *0, so the population is <u>increasing</u>, and as t -> ∞, y(t) will approach y = ∞. The limiting behavior is y(t) -> ∞.*

d. Use dsolve to find the solutions with initial values 1.5, 0.3, and 2.1.

Command Window Screenshot

Command Window

```
>> syms y(t)
>> dY = diff(y,t)

dY(t) =

diff(y(t), t)

>> ode = dY == -(2 - y)*y;
>> a = dsolve(ode, y(0) == 1.5);
>> b = dsolve(ode, y(0) == 0.3);
>> c = dsolve(ode, y(0) == 2.1);
>> disp(a)
2/(exp(2*t - log(3)) + 1)

>> disp(b)
2/(exp(2*t + log(17/3)) + 1)

>> disp(c)
-2/(exp(2*t - log(21)) - 1)
```

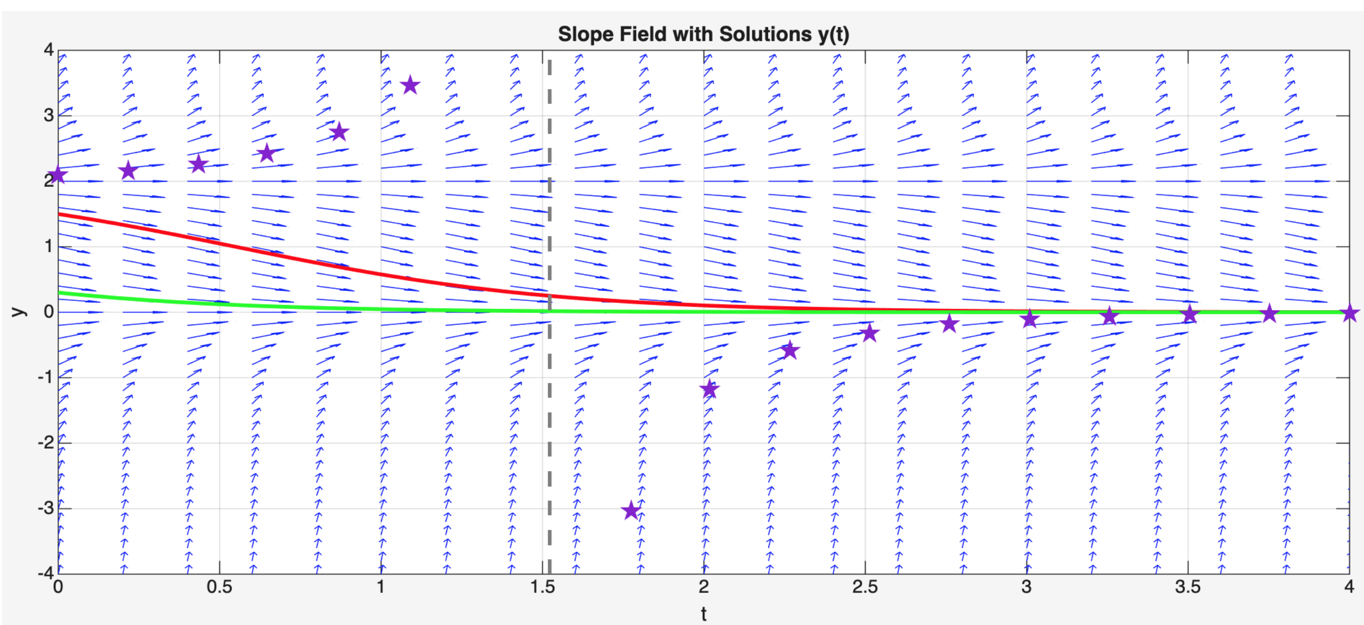
e. Plot the three solutions of part (d) together with the slope field on the same graph. Do the solutions follow the slope field as you expect it to? (You can use a for loop in this way: for cs = [1.5 0.3 2.1] (instructions to plot) end)

Command Window Screenshot

Command Window

```
>> [T, Y] = meshgrid(-4:0.2:4, -4:0.2:4);
>> dY = -(2 - Y).*Y;
>> A = sqrt(1 + dY.^2);
>> B = 1 ./ A;
>> C = dY ./ A;
>> figure;
>> quiver(T, Y, B, C, 0.5, 'b');
>> hold on
>> grid on
>> axis tight
>> xlabel('t')
>> ylabel('y')
>> title('Slope Field with Solutions y(t)');
>> syms y(t)
>> Dy = diff(y,t);
>> ode = Dy == -(2 - y)*y;
>> init_conds = [1.5 0.3 2.1];
>> solnColors = ['r','g','p'];
>> for i = 1:length(init_conds)
 y0 = init_conds(i);
sol = dsolve(ode, y(0) == y0);
fplot(sol, [0 4], solnColors(i), 'LineWidth', 2)
end
>> axis([0 4 -4 4])
```

Plot



% Legend: purple stars: y(0) = 2.1, red line: y(0) = 1.5, green line: y(0) = 0.3